School Worksheet

Class 12 Mathematics, Chapter 6: Application of Derivatives

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

A cylindrical tank is being filled with sand at the rate of $314\ \text{m}^3/\text{h}$. If the radius of the tank is 10 m, then the height of sand in the tank increases at the rate of: (Take $\pi = 3.14$)

(A) 1.1 m/h **(B)** 1 m/h

(C) π m/h **(D)** $\frac{\pi}{2}$ m/h

2 MCQ — 1 mark

The function $f(x) = x^3 - 3x^2 + 3x - 100$ is:

(A) increasing on $\mathbb{R}$ **(B)** decreasing on $\mathbb{R}$

(C) increasing on $(-\infty, 1)$ only **(D)** decreasing on $(1, \infty)$

3 MCQ — 1 mark

The interval in which $f(x) = x^2 - 4x + 6$ is strictly decreasing is:

(A) $(-\infty, 2)$ **(B)** $(2, \infty)$

(C) $(-\infty, \infty)$ **(D)** $(-2, 2)$

4 MCQ — 1 mark

The maximum value of $\sin x \cos x$ is:

(A) 1 **(B)** $\frac{1}{2}$

(C) 2 **(D)** $\frac{1}{\sqrt{2}}$

5 MCQ — 1 mark

The local minimum value of $f(x) = x + \frac{1}{x}$, $x > 0$, is:

(A) 1 **(B)** -2

(C) 2 **(D)** 0

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** $f(x) = \sin x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$. **Reason (R):** If $f'(x) > 0$ for every x in an open interval, then f is strictly increasing in that interval. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer The edge of a cube is increasing at the rate of 3 cm/s. How fast is the volume increasing when the edge is 10 cm long?	2 marks

Level 2

19 marks

8	Short answer Find the intervals in which $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) strictly increasing (b) strictly decreasing.	3 marks
9	Short answer Air is pumped into a spherical balloon at the rate of $900\ \text{cm}^3/\text{s}$. Find the rate at which the radius is increasing when the radius is 15 cm.	2 marks
10	Short answer Find the local maximum and local minimum values of $f(x) = x^3 - 6x^2 + 9x + 15$.	3 marks
11	Short answer Find the intervals in which the function $f(x) = \frac{\log x}{x}$ is strictly increasing or strictly decreasing.	3 marks
12	Short answer A ladder 5 m long leans against a wall. The bottom of the ladder is pulled along the ground away from the wall at 2 cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4 m from the wall?	3 marks
13	Short answer Find the absolute maximum and absolute minimum values of $f(x) = \sin x + \cos x$ on $[0, \pi]$.	3 marks
14	Short answer Find two positive numbers whose sum is 15 and the sum of whose squares is minimum.	2 marks

Level 3

25 marks

15	Long answer Show that of all rectangles inscribed in a given fixed circle, the square has the maximum area.	5 marks

Q No.	Question	Marks
16	Long answer Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume.	5 marks
17	Long answer A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window so as to admit maximum light through the whole opening.	5 marks
18	Short answer Find the intervals in which $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$, is strictly increasing or strictly decreasing.	3 marks
19	Short answer A man 2 m tall walks at a uniform speed of 5 km/h away from a lamp post 6 m high. Find the rate at which the length of his shadow increases.	3 marks
20	Case study **Case Study:** A technical company is designing a rectangular solar panel installation on a roof using 300 metres of boundary material. The design includes a partition running parallel to one of the sides, dividing the area into two sections. Let the length of the side perpendicular to the partition be x metres and the side parallel to the partition be y metres. **(i)** Write the equation for the total boundary material used (boundary and partition) in terms of x and y. — 1 mark **(ii)** Write the area of the solar panel as a function of x. — 1 mark **(iii)(a)** Find the critical point of the area function and use the second derivative test to find the maximum area. — 2 marks OR **(iii)(b)** Using the first derivative test, find the maximum area the company can enclose with the 300 m of boundary material. — 2 marks	4 marks

Answer key

Chapter 6: Application of Derivatives, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) 1 m/h
2	(A) increasing on $\mathbb{R}$
3	(A) $(-\infty, 2)$
4	(B) $\frac{1}{2}$
5	(C) 2
6	(A) Both A and R are true, and R is the correct explanation of A.
7	The volume is increasing at $900\ \text{cm}^3/\text{s}$.

Level 2

8	(a) Strictly increasing on $(-\infty, -2) \cup (3, \infty)$; (b) strictly decreasing on $(-2, 3)$.
9	The radius is increasing at $\dfrac{1}{\pi}$ cm/s.
10	Local maximum value $= 19$ (at $x = 1$); local minimum value $= 15$ (at $x = 3$).
11	Strictly increasing on $(0, e)$; strictly decreasing on (e, ∞).
12	The height on the wall is decreasing at $\dfrac{8}{3}$ cm/s.
13	Absolute maximum $= \sqrt{2}$ (at $x = \frac{\pi}{4}$); absolute minimum $= -1$ (at $x = \pi$).
14	The numbers are $\dfrac{15}{2}$ and $\dfrac{15}{2}$.

Level 3

15	Area is maximum when $x = y = \sqrt{2}\,a$, i.e. the rectangle is a square.
16	Height $= \dfrac{2R}{\sqrt{3}}$; maximum volume $= \dfrac{4\pi R^3}{3\sqrt{3}}$ cubic units.
17	Width $= \dfrac{20}{\pi + 4}$ m, height of rectangle $= \dfrac{10}{\pi + 4}$ m (radius of semicircle $= \dfrac{10}{\pi + 4}$ m).
18	Strictly increasing on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; strictly decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.
19	The shadow increases at $\dfrac{5}{2}$ km/h.
20	(i) $2x + 3y = 300$; (ii) $A(x) = 100x - \dfrac{2x^2}{3}$; (iii) maximum area $= 3750\ \text{m}^2$ at $x = 75$ m, $y = 50$ m.